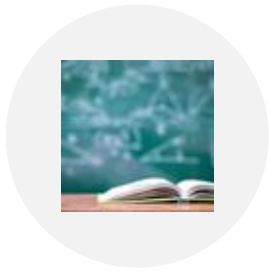


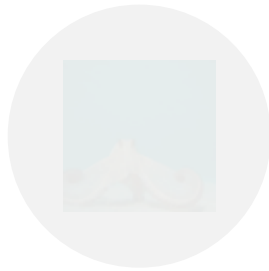
DATASCI 185: Computational Methods for Data Science

Dr. Peter Sentz
Dr. Nathan Gabriel

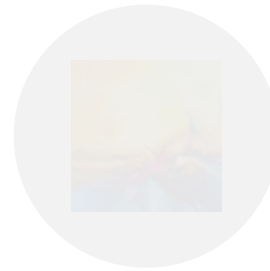
Agenda



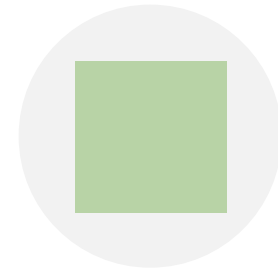
INTRODUCTIONS



CLASS DETAILS



COMPUTING
ENVIRONMENT



INSTALLATION



About me

- My name is Peter Sentz
 - Calling me “Peter” is fine
 - If you’re too scared to do that, “Prof. Sentz” or “Dr. Sentz” works as well.
- Academic Background
 - Studied Mathematics (minoring in Economics) at University of Wisconsin-Milwaukee
 - Master’s degree in Applied Mathematics at University of Washington
 - PhD in Computer Science at University of Illinois Urbana-Champaign
 - Postdoc in Applied Mathematics at Brown University.
 - This is my second semester at Emory University
- Pictured: my (now 14 month old) daughter
- My research background is in the numerical solution of differential equations and scientific machine learning.

About me



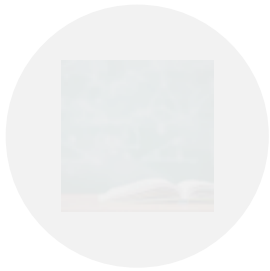
- My name is Nathan Gabriel
 - You can call me Professor Nathan, Professor Gabriel, Dr. Gabriel or just Professor (when you forget my name)
- Academic Background
 - Studied Mathematics at Rice University
 - Master's degree in Philosophy at University of Houston
 - Master's degree in Social Science at University of California, Irvine
 - PhD in Logic and Philosophy of Science at University of California, Irvine
 - Postdoc Cognitive & Information Sciences at University of California Merced
- My research background is in the computational approaches to network science and evolutionary game theory, with an emphasis on cultural evolution.

Teaching Assistants

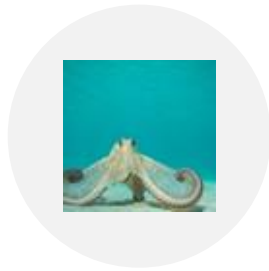
- Maggie Dai
- Tucker Sampson

They will hold office hours throughout the semester (the schedule is TBD).

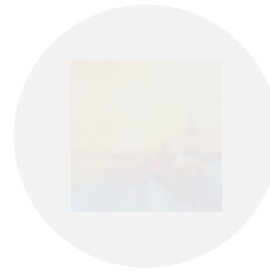
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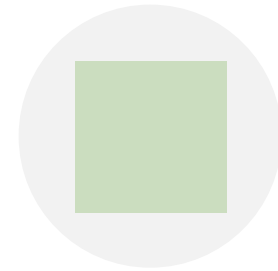
INTRODUCTIONS



CLASS DETAILS



COMPUTING
ENVIRONMENT



INSTALLATION

Learning Objectives

- Understand basics of GitHub version control
- Learn about key programming principles in Python
- Learn how to manipulate and analyze data in the Pandas library
- Communicate results using plots and other data visualization

For more details and class schedule, review syllabus on course Canvas page!

Grades

- **Assignments (x 10): 50%**
- **Quizzes (x5): 30%**
- **Final Project: 20 %**
 - Will provide details mid-semester
 - Due at the end of the semester (December 9)
 - Groups of 3-4 students
- **Midsemester Survey: + 0.5% (Extra)**
- **Final Course Evaluations: + 0.5% (Extra)**

Late submissions

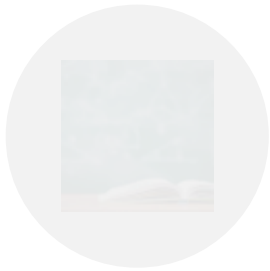
- Late assignments will receive a 5% penalty for every 12 hour period after the due date. After 5 days, late assignments will receive no credit.
- To account for unforeseen circumstances, **we will drop the lowest assignment and the lowest quiz**
- The first assignment covers the basics of the software we will use. You will need these to be able to use the lectures notes.

Coding ability is not innate

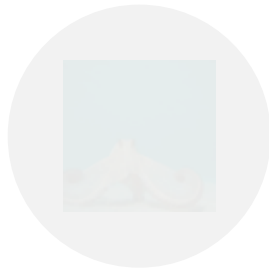
- Coding ability can be developed.
- Academic skills and abilities are acquired through hard work, **mistakes**, and perseverance.
- This class is an introductory programming – using AI/LLM tools is **not appropriate** at this stage (and not allowed).
- My only goal here is that you learn the material. Please ask me questions!

Questions about the logistics

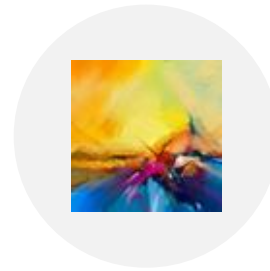
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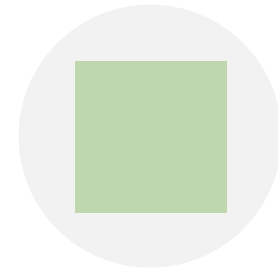
INTRODUCTIONS



CLASS DETAILS



**COMPUTING
ENVIRONMENT**



INSTALLATION

Two components:



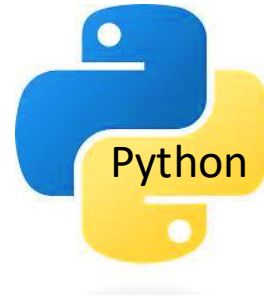
ANACONDA

Virtual Environment



VS Code
Integrated
Development
Environment (IDE)

+



(in background)

Programming language we use to
write the instructions

The “front-end” software that
you open every time you code
something in Python

A “back-end” software that
you install once and hardly
open again.

We open script files in VSCode



Script* File

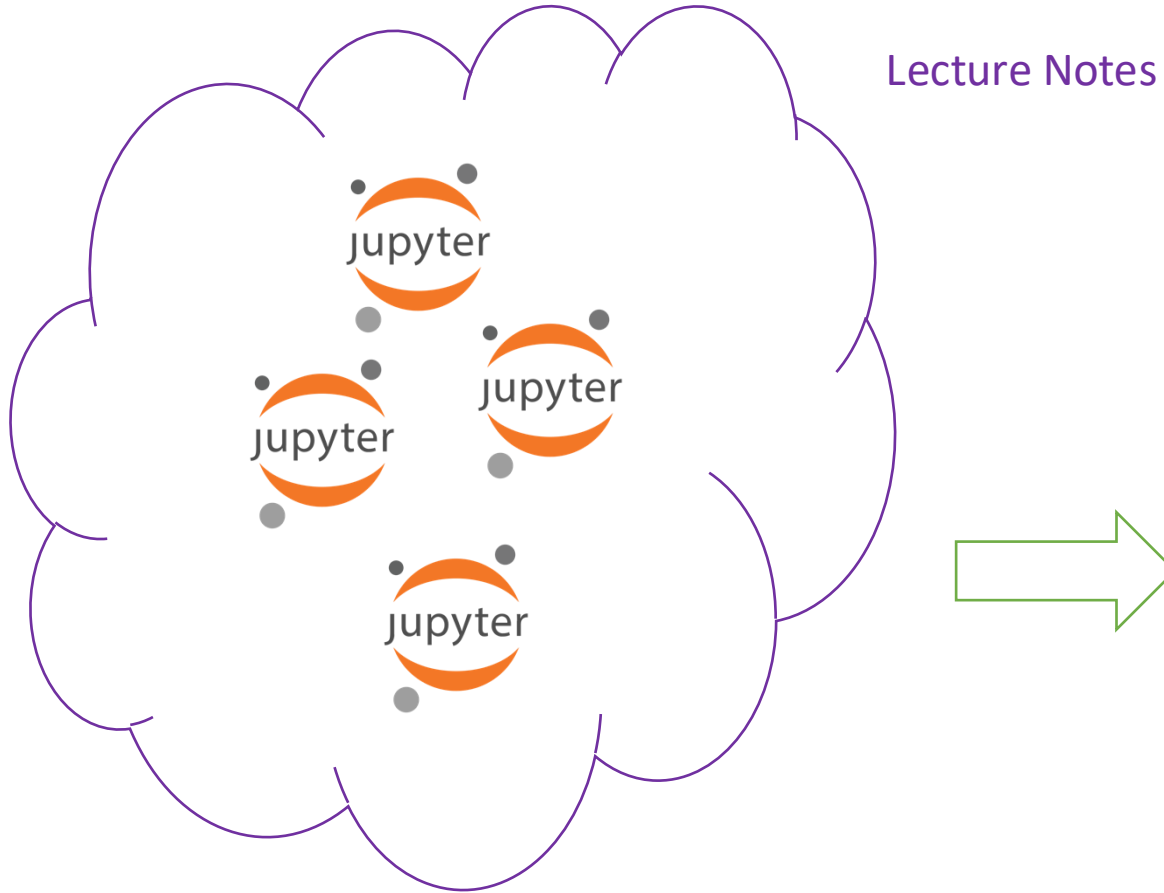
- File with code in the Python **programming language**.
- Instructions for the program to follow



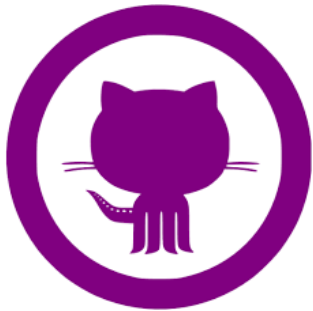
VS Code Integrated Development Environment (IDE)

Environment where the user
writes the scripts

* Not really script files. Jupyter Notebook files



VS Code

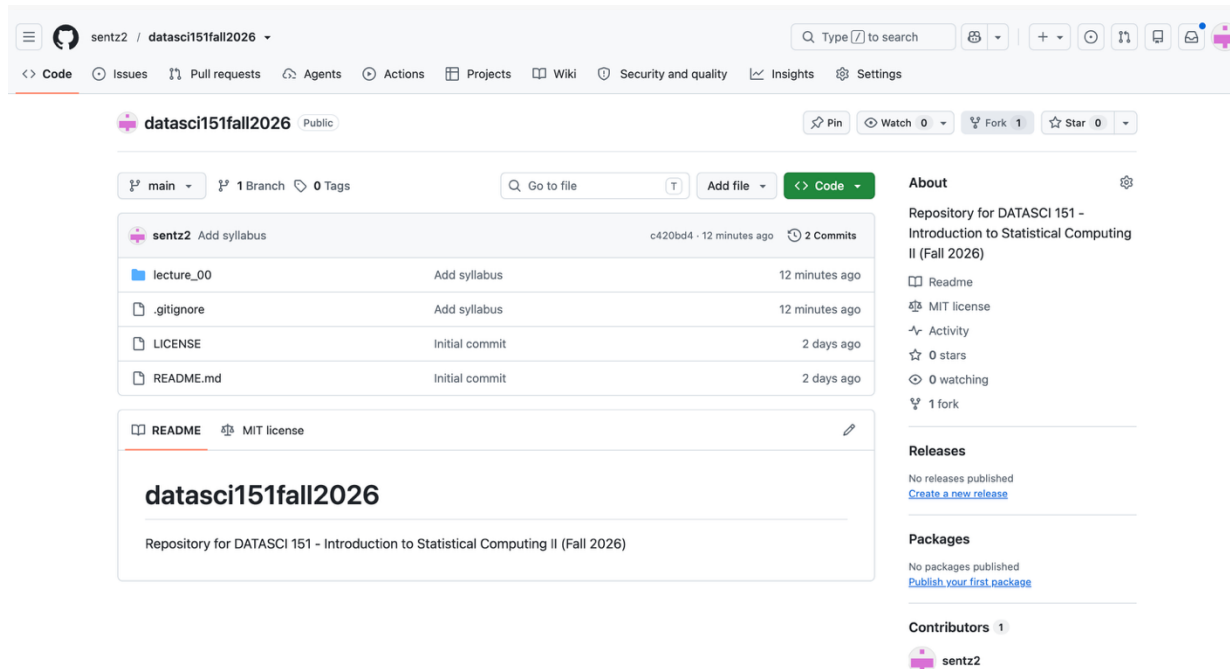


Github:
A file management system in
the cloud (with desktop app)

- Has version control
- Great for collaborative programming

Lecture notes are publicly available at my GitHub website (link posted on Canvas):

<https://github.com/sentz2/datasci151fall2026>

A screenshot of a GitHub repository page for 'sentz2 / datasci151fall2026'. The repository is public and has 1 fork and 0 stars. The main branch is 'main'. The repository contains files: 'lecture_00', '.gitignore', 'LICENSE', and 'README.md'. The README file is selected and shows the title 'datasci151fall2026' and the description 'Repository for DATASCI 151 - Introduction to Statistical Computing II (Fall 2026)'. The right sidebar shows the 'About' section with the repository description, 'Readme', 'MIT license', 'Activity', '0 stars', '0 watching', and '1 fork'. There are also sections for 'Releases' and 'Packages'.

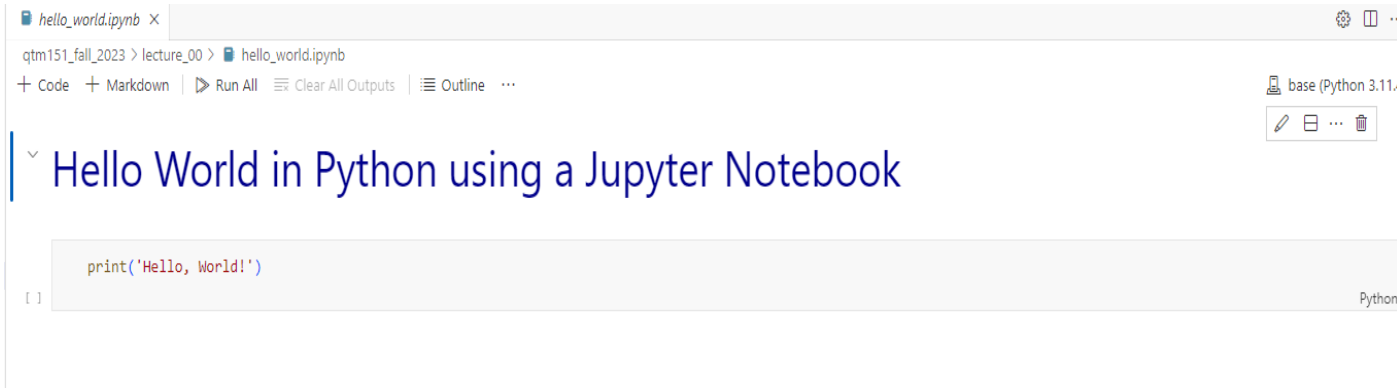
You will be able to view the lectures in the browser

A screenshot of a Jupyter Notebook code cell. The cell title is 'Hello World in Python using a Jupyter Notebook'. The code cell contains the following Python code:

```
In [ ]: print('Hello, World!')
```



A Jupyter Notebook (“.ipynb”) is a file with code (python) and annotations (markdown)



- All the lecture notes are written as Jupyter notebooks
- It is encouraged that you bring your laptop to class
- Lecture notes are designed to be follow-along. There will be “try it yourself” exercises.

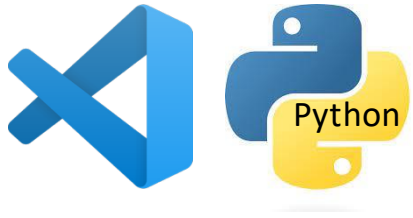


VS Code

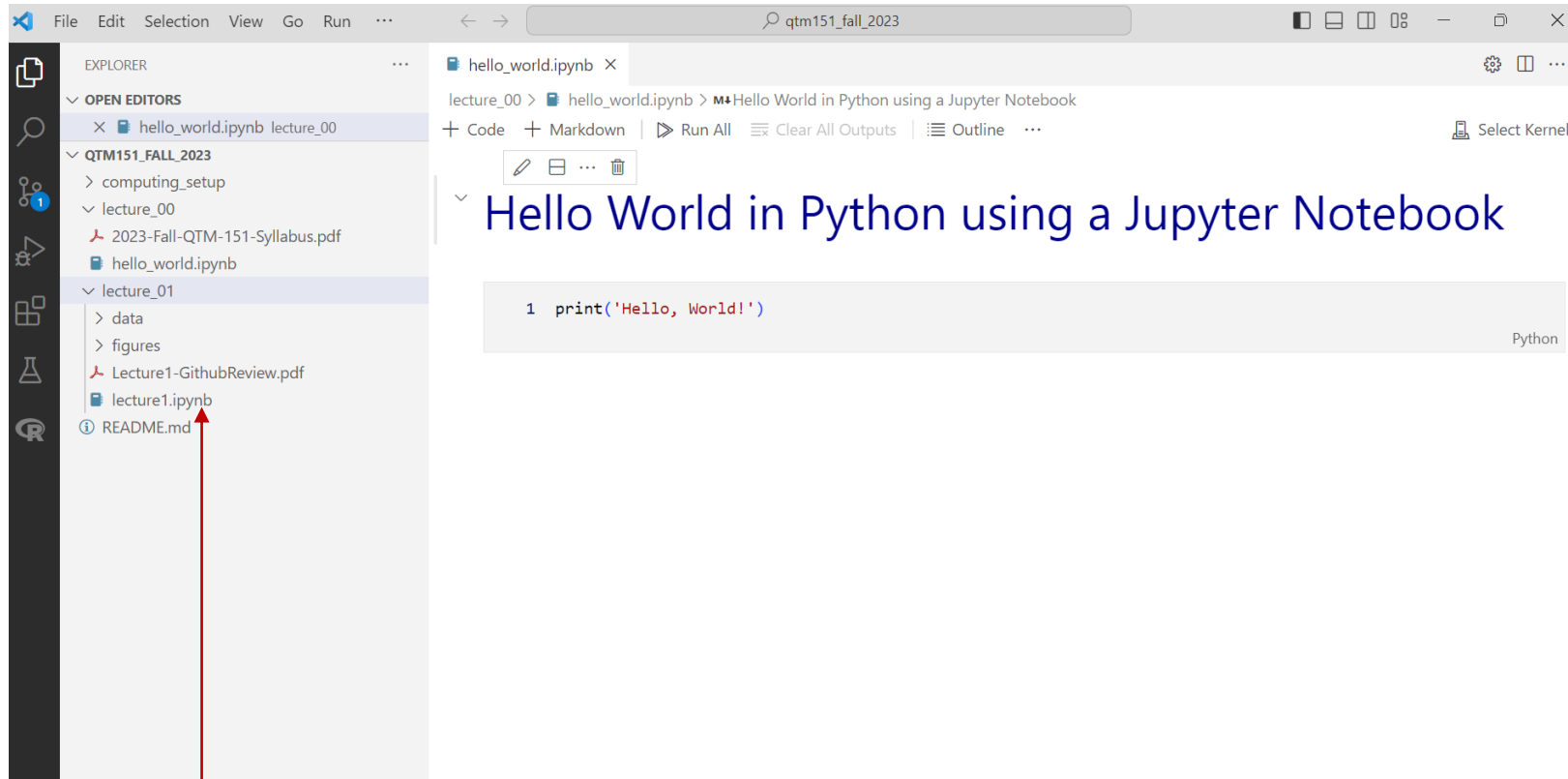
See **Assignment 1** (available on Gradescope tab in Canvas)

Basics of a Jupyter notebook

- Need to have all the tools installed
- Install soon that we can help you with any issues!



We will do all our coding in Visual Studio Code



Lecture notes

“.ipynb”: Interactive Python Notebook

Github Desktop will allow you to **automatically download the lectures notes** from my account (and update them).

- It will also show you
- the date of any update
 - what the changes are

Update the lecture notes before class!

Current Repository
qtm151spring2025

Current Branch
main

Fetch origin
Last fetched 24 minutes ago

Changes 20

History

No Branches to Compare

Upload first few lecture notes
Alejandro Sanchez-Becerra • 26 minutes ago

Update README.md
Alejandro Sanchez-Becerra • 42 minutes ago

Initial commit
Alejandro Sanchez-Becerra • 42 minutes ago

Update README.md

Alejandro Sanchez-Becerra 74f51d4 +11 -1

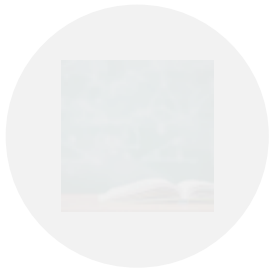
1 changed file

README.md

README.md	
@@ -1	
1	- # qtm1
	1 + # QTM
2	+
3	+ This c
	nts wi
	studen
4	+
5	+ ## Bas
6	+
7	+ Go to
8	+
9	+ - Inst
10	+ - Inst
11	+ - Gith

Questions about computing environment?

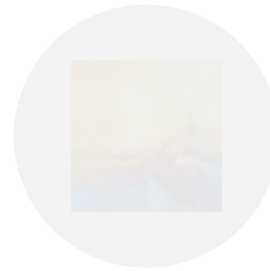
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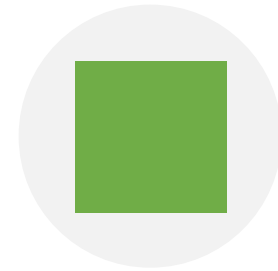
INTRODUCTIONS



CLASS DETAILS



COMPUTING
ENVIRONMENT



INSTALLATION

Let's get started!

Go to “Modules” in Canvas and open the documents under “Installation” and follow instructions.

- I will stick around to help if you want to try installing everything today
- If you don't complete the installation, we will briefly return to this next class.
- Please reach out on the “Discussions” tab in Canvas, or email me if you want to schedule office hours before they are scheduled.
- A poll for office hours times will be released later today. Keep an eye out for Canvas announcements!